

muscular activity, signals that most conventional sensors cannot detect. Combine that sensitivity with their small size and scalability, and TMR sensors become strong candidates for future quantum computing systems once paired with quantum hardware developers.

The race to what's next

When we look at who is leading in TMR sensor technology, we see different companies driving progress in different ways. Allegro MicroSystems, for example, has shown how much noise performance can be improved. Their XtremeSense TMR sensors cut noise levels compared to both earlier TMR devices and Hall-based solutions, which translates into more efficient power conversion. For us, that matters in real systems like solar inverters, heat pumps, and EV chargers, where every bit of efficiency adds up.

TDK Corporation offers another perspective on what TMR sensors can achieve. Their 360° angle sensor demonstrates how this technology can precisely measure rotor position in BLDC motors. These are motors we depend on in safety-critical applications such as traction motors, brake boosters, and power steering. What stands out is the sensor's ability to remain accurate across a wide range of temperatures and over time, while still operating reliably in extended magnetic-field ranges. It reminds us that TMR is not just about sensitivity; it is also about stability and trust in challenging environments.

And when it comes to pushing boundaries, Neuramics is taking us into uncharted territory: picoTesla sensitivity. While most commercial sensors stop at the nanoTesla range, they are working on detection down to 1 picoTesla per $\sqrt{\text{Hz}}$ at 1Hz. They have already demonstrated a four-channel system that

operates effectively even in noisy magnetic environments. This kind of sensitivity opens up entirely new areas to explore—whether in monitoring cardiac signals for healthcare, advancing quantum computing, or applying the technology in agriculture and robotics. Their emphasis on sensitivity, stability, smaller dimensions, and low power underlines what it takes to make these advances practical.

Key leaders in the high-sensitivity TMR sensing space include the stated examples, while other major magnetic sensor manufacturers, such as Infineon Technologies, Honeywell, and STMicroelectronics, remain active in the broader market. This shows that innovation is happening at multiple levels, with some companies driving breakthroughs in sensitivity and performance, and others building reliable solutions for established applications.

TMR sensors are no longer laboratory curiosities. They are becoming practical tools that enable us to measure and interact with the world in ways that were once impossible. We can now detect faint heart signals, track gestures, enhance power system efficiency, and even contribute to quantum technologies. As we refine their sensitivity, shrink their size, and reduce power consumption, these sensors are transitioning from experimental setups to everyday devices. The journey is still unfolding, but one thing is clear: with TMR sensors, we are entering a future where invisible magnetic fields reveal powerful new insights and possibilities. **EFY**

This article is based on a conversation with Kia Nazarpour, Chief Strategy Officer at Neuramics, Professor of Digital Health at the University of Edinburgh, with experience in industry R&D, and co-founder of Rebel Bionics, which develops prosthetics for children. It is written and curated by Nidhi Agarwal.

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